Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 09
Chapter 13: Statistics

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If the arithmetic mean of $7, 8, x, 11, 14$ is x, then the value of x is: [1]
(A) 9 (B) 9.5 (C) 10 (D) 10.5

Q2. The difference between the upper limit of the median class and the lower limit of the modal class for the data is: [1]

Class	0–5	5–10	10–15	15–20	20–25
Frequency	10	15	12	20	9

(A) 0 (B) 5 (C) 10 (D) 15

Q3. If a variable takes values $0, 1, 2, \ldots, n$ with frequencies proportional to $1, 1, 1, \ldots, 1$, then its mean is: [1]
(A) $n/2$ (B) $(n+1)/2$ (C) $n(n+1)/2$ (D) n

Q4. For a symmetrical frequency distribution, which of the following relations is correct? [1]
(A) Mean > Median > Mode
(B) Mean < Median < Mode
(C) Mean = Median = Mode
(D) Mode = $\frac{\text{Mean}+\text{Median}}{2}$

Q5. If the mode of a frequency distribution is 55 and the median is 50, then its mean is: [1]
(A) 47.5 (B) 48.0 (C) 52.5 (D) 45.0

Q6. **Assertion (A):** The cumulative frequency of the last class interval in a 'less than' cumulative distribution is equal to the total frequency N.
Reason (R): Cumulative frequency of a class is the sum of frequencies of that class and all preceding classes. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. In a continuous distribution, $l = 40, h = 10, f_1 = 20, f_0 = 12, f_2 = 11$. Calculate the mode. [2]

Q8. In a grouped distribution, $l = 25, N = 50, cf = 18, f = 14, h = 5$. Calculate the median. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Find the median of the distribution of letters in surnames: [3]

Letters	1–4	4–7	7–10	10–13	13–16	16–19
Surnames	6	30	40	16	4	4

Q10. The mean of the following distribution is 25. Find the value of p: [3]

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	18	15	p	6

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

Q11. A life insurance agent found the following data for distribution of ages of 100 policy holders. Calculate the median age and the mean age: [5]

Age (in years)	Below 20	Below 25	Below 30	Below 35	Below 40
Number of policy holders	2	6	24	45	78

Age (in years)	Below 45	Below 50	Below 55	Below 60	–
Number of policy holders	89	92	98	100	–

Q12. If the mean of the following frequency distribution is 145 and the total frequency is 100, find the missing frequencies f_1 and f_2: [5]

Class	100–120	120–140	140–160	160–180	180–200
Frequency	10	f_1	30	f_2	15

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: NATIONAL EXPRESS HIGHWAY SMART FASTAG TOLL CLEARANCE EFFICIENCY [4 Marks]

The National Highways Authority analyzed vehicle transit processing delays (in minutes) at an interstate electronic toll plaza during holiday weekend congestion.

Transit Delay (minutes)	0–5	5–10	10–15	15–20	20–25
Vehicles Logged	15	35	50	30	20

(i) Calculate the modal delay time experienced by vehicles at the toll plaza. [2]

(ii) What is the total number of vehicles surveyed in this clearance audit? [1]

(iii) Find the median class of the delay distribution. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★